Literature Review: Love of Variety

Tate Mason

John Munro Godfrey Sr. Dept. of Economics
University of Georgia

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Roadmap

1. Motivation
2. Variety-Seeking & Habit Formation
3. Discrete-Choice Demand Estimation
4. Dynamic Advertising
5. Summary

Motivation

The problem with standard logit:

- A new product always raises utility via a fresh ε draw
- No actual parameter for variety-seeking
- Consumers seem to prefer goods that are *novel but familiar*

This project models:

- How deviation from a consumer's *experienced mean* affects utility
- How advertising interacts with variety

Three Literatures

1. Variety-seeking & habit formation
2. Dyanmic discrete-choice demand estimation
3. Dynamic advertising

Data

Nielsen-Kilts Homescan and Marketscan scanner data

Roadmap

Pollak (1970): Habit Formation

Citation

Pollak, R. A. (1970). Habit formation and dynamic demand functions. *Journal of Political Economy*, 78(4), 745–763.

What they do:

- Model consumer preferences as influenced by past consumption
- Show conditions under which consumers derive utility from previously consumed goods
- Theoretical foundation for state-dependence in demand

Relevance: Seminal theoretical grounding for the idea that purchase history shapes current period choices — the core mechanism of this project

Dixit & Stiglitz (1977): Love of Variety

Citation

Dixit, A. K. and Stiglitz, J. E. (1977). Monopolistic competition and optimum product diversity. *American Economic Review*, 67(3), 297–308.

What they do:

- Model preferences over a continuum of differentiated products
- Show consumers prefer greater variety under monopolistic competition
- Derive optimal product diversity conditions

Relevance: The canonical "love of variety" framework — provides the theoretical counterpart to Pollak's habit formation, motivating a model that nests both

McAllister & Pessemier (1982): Variety-Seeking Review

Citation

McAllister, I. and Pessemier, E. A. (1982). Variety seeking behavior: An interdisciplinary review. *Journal of Consumer Research*, 9(3), 311–322.

What they do:

- Comprehensive review of variety-seeking literature through early 1980s
- Synthesize psychological, sociological, and economic perspectives
- Catalog empirical patterns in variety-seeking behavior

Relevance: Foundational reference for understanding motivations behind variety-seeking; useful for grounding the model's behavioral assumptions

Becker & Murphy (1988): Rational Addiction

Citation

Becker, G. S. and Murphy, K. M. (1988). A theory of rational addiction. *Journal of Political Economy*, 96(4), 675–700.

What they do:

- Model addictive consumption as rational forward-looking behavior
- Past consumption raises the marginal utility of current consumption
- Embed habit stock dynamics into utility maximization

Relevance: Habit-stock approach maps onto the experienced-mean state variable $\bar{X}_{it}$; provides microfoundations for why $\bar{X}_{it}$ is welfare-relevant

Bronnenberg, Dubé & Gentzkow (2021): Brand Preferences

Citation

Bronnenberg, B. J., Dubé, J.-P., and Gentzkow, M. (2021). Millenials and the Take-off of Craft Brands: Preference Formation in the U.S. Beer Industry. *Working Paper*.

What they do:

- Use consumer migration across markets as identification strategy
- Examine craft beer industry; find strong persistence in brand preferences
- Construct a "capital stock" measure to track consumption history

Relevance: Capital stock $\approx$ this project's $\bar{X}_{it}$; inspires the "prior mean" initialization in simulations. Key empirical motivation for persistence

Roadmap

McFadden (1974): Conditional Logit

Citation

McFadden, D. (1974). Conditional logit analysis of qualitative choice behavior. In Zarembka, P. (ed.), *Frontiers in Econometrics*. Academic Press.

What they do:

- Derive the multinomial logit from random utility maximization
- Show IIA property and its implications
- Establish the statistical framework for discrete choice

Relevance: The MNL is the baseline for this project's choice probabilities. The $\varepsilon_{ijt} \sim$ Gumbel assumption driving P_{ijt} comes directly from this framework

Berry, Levinsohn & Pakes (1995): BLP

Citation

Berry, S., Levinsohn, J., and Pakes, A. (1995). Automobile prices in market equilibrium. *Econometrica*, 63(4), 841–890.

What they do:

- Introduce random-coefficients logit for differentiated products
- Solve the market-share inversion problem via contraction mapping
- Provide IV strategy to handle price endogeneity

Relevance: The estimation framework for this project. BLP instruments will identify preferences for sameness and variety from observed market shares; random coefficients accommodate heterogeneity in variety preferences

Nevo (2001): BLP Applied to Cereal

Citation

Nevo, A. (2001). Measuring market power in the ready-to-eat cereal industry. *Econometrica*, 69(2), 307–342.

What they do:

- Apply BLP to the ready-to-eat cereal industry
- Carefully construct instruments; demonstrate identification
- Recover markups and market power estimates

Relevance: Practical implementation guide for BLP; cereal is a close cousin to the CPG scanner data setting. Blueprint for the estimation phase of this project

Conlon & Gortmaker (2020): PyBLP Best Practices

Citation

Conlon, C. and Gortmaker, J. (2020). Best practices for differentiated products demand estimation with PyBLP. *RAND Journal of Economics*, 51(4), 1108–1161.

What they do:

- Provide numerical best practices for BLP-style estimators
- Document common pitfalls (integration rules, convergence tolerance)
- Release PyBLP, an open-source Python implementation

Relevance: Directly relevant for the structural estimation phase. PyBLP will be the estimation engine; this paper gives guidance on numerical implementation choices

Wei & Jiang (2024): Neural Network Estimation

Citation

Wei, Y. and Jiang, Z. (2024). Estimating parameters of structural models using neural networks. *Marketing Science*, 44(1), 102–128.

What they do:

- Implement neural networks as a flexible moment-matching estimator
- Demonstrate efficiency gains over classical GMM in structural models
- Show improved accuracy given correct moment selection

Relevance: Potential computational shortcut for estimating the dynamic model on Nielsen-Kilts data; ties into Song (2025) as a practical implementation strategy

Roadmap

Nerlove & Arrow (1962): Goodwill

Citation

Nerlove, M. and Arrow, K. J. (1962). Optimal advertising policy under dynamic conditions. *Economica*, 29(114), 129–142.

What they do:

- Model advertising as a stock of "goodwill" that depreciates over time
- Derive the firm's optimal advertising path as a dynamic control problem
- Show advertising investment trades off current cost vs. future demand

Relevance: Theoretical foundation for the firm's dynamic advertising problem. The goodwill stock maps onto consumer familiarity, linking naturally to the habit formation side

Erdem & Keane (1996): Advertising as Learning

Citation

Erdem, T. and Keane, M. P. (1996). Decision-making under uncertainty: Capturing dynamic brand choice processes in turbulent consumer goods markets. *Marketing Science*, 15(1), 1–20.

What they do:

- Model advertising as a signal that updates consumer beliefs about quality
- Consumers Bayesian-update perceptions from ad exposure
- Estimate dynamic brand choice model on detergent data

Relevance: Estimation procedure and "perceptions formed via advertising" are directly relevant. This project *flips* the direction: consumer choices inform firm advertising strategy

Dubé, Hitsch & Rossi (2005): Adverising Timing

Citation

Dubé, J.-P., Hitsch, G. J., and Rossi, P. E. (2005). An Empirical Model of Adverising Dynamics. *Quantitative Marketing and Economics*, 3, 107-144.

What they do:

- Utilize goodwill stock with distributed lag to model advertising as investment process
- Solve for Markov perfect equilibrium in a dynamic oligopoly game
- Using frozen entree data, find that their model overpredicts advertising persistence

Relevance: Update on the Nerlove-Arrow framework; the distributed lag structure informs the specification of the firm's advertising cost function.

Bergemann & Bonatti (2011): Optimal Targeting

Citation

Bergemann, D. and Bonatti, A. (2011). Targeting in advertising markets: Implications for offline vs. online media. *RAND Journal of Economics*, 42(3), 417–443.

What they do:

- Model optimal targeting in a static advertising market
- Firm chooses which consumer segments to target and how intensively
- Compare offline (coarse) vs. online (fine) targeting technologies

Relevance: Starting point for the dynamic targeting extension. This project extends to a dynamic setting where targeting depends on the consumer's $\bar{X}_{it}$ state

Hosanagar, Fleder, Lee & Buja (2014): Recommender Systems

Citation

Hosanagar, K., Fleder, D., Lee, D., and Buja, A. (2014). Will the global village fracture into tribes? Recommender systems and their effects on consumer fragmentation. *Management Science*, 60(4), 805–823.

What they do:

- Ask whether algorithmic recommendations fragment or homogenize consumption
- Construct attribute-distance measures across product pairs
- Find recommendations *homogenize* consumption despite personalization

Relevance: Attribute distance measures are repurposed here on the *firm* side to determine when and whom to advertise to; homogenization result informs expected equilibrium outcomes

Song (2025): Personalization Field Experiment

Citation

Song, Yu. (2025). Optimizing multi-stage personalization in the customer journey. *Working Paper*.

What they do:

- Partner with a major e-tailer for a randomized field experiment
- Vary personalization intensity across stages of the consumer search journey
- Identify causal effect of personalization on consumption patterns

Relevance: Practical implementation of Wei & Jiang (2024) neural methods in a live commercial setting; provides external validity for the structural model's policy predictions

Roadmap

How the Papers Fit Together

Consumer side

- Pollak (1970): habit formation microfoundations
- Dixit-Stiglitz (1977): love of variety
- McAllister-Pessemier (1982): behavioral evidence
- Becker-Murphy (1988): rational addiction stock
- Bronnenberg et al. (2012): empirical persistence

Firm / advertising side

- Nerlove-Arrow (1962): goodwill dynamics
- Erdem-Keane (1996): learning from ads
- Dube et al. (2005): ad timing and persistence
- Bergemann-Bonatti (2011): optimal targeting
- Hosanagar et al. (2014): fragmentation vs. homogenization
- Song (2025): field experiment, neural estimation

Contribution of This Paper

Where This Project Sits

We connect the love-of-variety / habit-formation demand literature to the dynamic targeting literature by making $\bar{X}_{it}$ the **sufficient statistic** for both consumer switching behavior and firm advertising decisions.

1. **Novel mechanism:** Utility depends on deviation from experienced mean — nests Pollak and Dixit-Stiglitz in a single dynamic framework
2. **Dynamic targeting:** Extends Bergemann-Bonatti to a setting where the firm's optimal ad policy is a function of the consumer's history
3. **Estimation:** Bring the model to Nielsen scanner data using BLP/PyBLP with potential neural-network moments